Computer Graphics

Week 2 : *Digital Differential Analyzer*

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OUTLINES

1. Line Equation
2. Digital Differential Analyzer (DDA)

1. Line Equation

Line Formation Algorithm

The line equation according to Cartesian coordinates is:

$y = mx + B$ (explicit equation)

Where gradient m is the slope of the line formed by two points of ($x1$, $y1$) and ($x2$, $y2$). It is said to be explicit equation because there is a symbol m in the formula.

Substitute $x = a$ and $y = b$ in the equation line $y = mx + B$ so that it is obtained:

$b = ma + B$

or

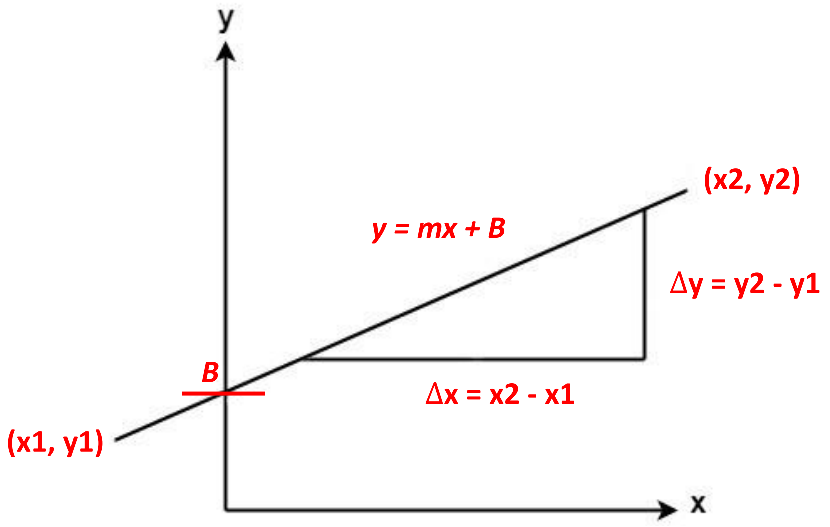
$\boldsymbol{B = b - ma}$

Substitute the value of $\boldsymbol{B}$ in the initial equation, $y = mx + \boldsymbol{B}$ so that it is obtained:

$y = mx + (\boldsymbol{b - ma})$

$y - b = mx - ma$

$y - b = m(x - a)$



So, the equation of the line that passes the point (a, b) with the gradient m is $\boldsymbol{y - b = m(x - a)}$

Line Formation Algorithm (2)

if we find the the line equation $ax + by = c$ (implicit equation), to find $\boldsymbol{m}$ we can use the following method:

$$\boldsymbol{m} = -\frac{coefficient\ x}{coefficient\ y}$$

Example : $x - 3y = -6$

$$\boldsymbol{m} = -\frac{coefficient\ x}{coefficient\ y}$$

$$\boldsymbol{m} = -\frac{1}{-3} = \frac{\mathbf{1}}{\mathbf{3}}$$

Relating Explicit to Implicit Equation

explicit → *y = mx + B*

implicit → *F(x,y) = ax + by + c = 0*

Hence,

$$y = \left(\frac{dy}{dx}\right)x + B$$

$$\left(\frac{dy}{dx}\right)x - y + B = 0$$

$$F(x,y) = (dy)x + (-dx)y + (dx)B = 0$$

Where,

$$a = (dy);\ b = -(dx);\ c = (dx)B$$

Example 1

Determine the equation of the line through point (-4, 5) with gradient 2!

Answer:

$a = -4;$

$b = 5;$

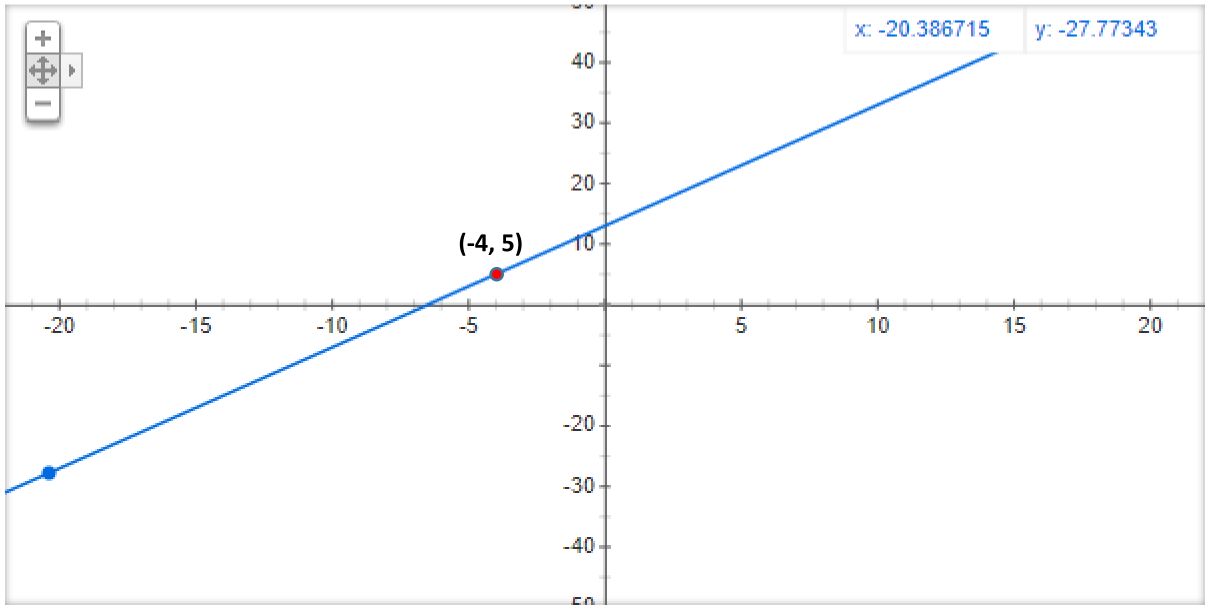
$m = 2$

$y - b = m\,(x - a) \quad \Leftrightarrow \quad y - 5 = 2\,(x - (-4))$

$\Leftrightarrow y - 5 = 2\,(x + 4)$

$\Leftrightarrow y - 5 = 2x + 8$

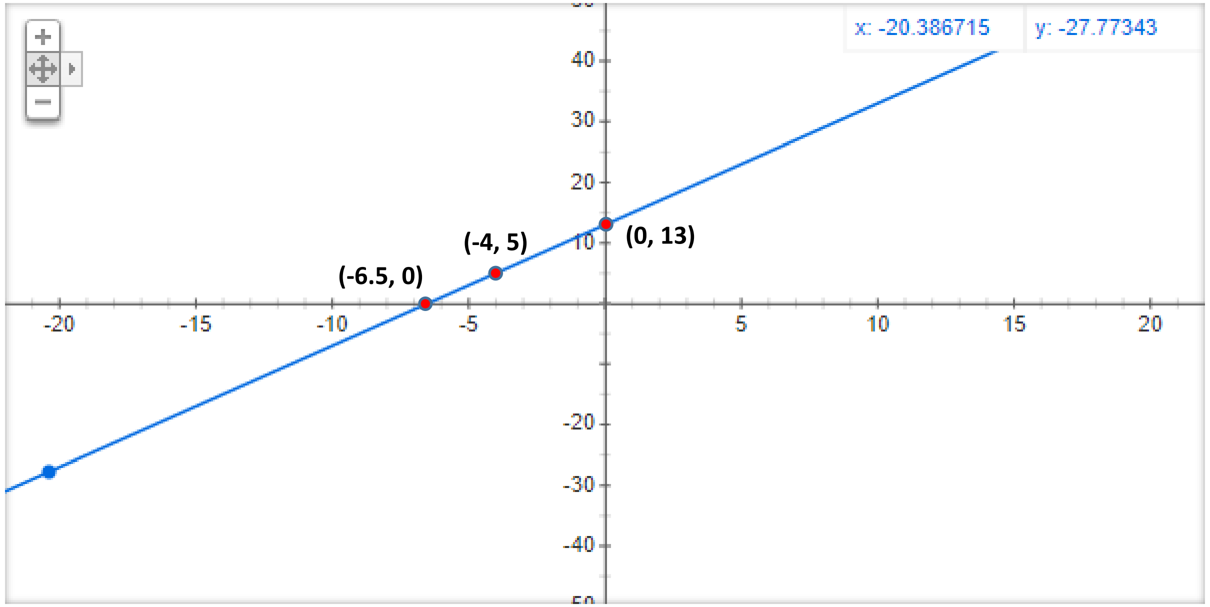
$\Leftrightarrow y = 2x + 13$



How to Draw Lines

At the equation *$y = 2x + 13$*, if we choose *$x = 0$*, we get *$y = 13$*. if we choose *$y = 0$*, we get *$x = –6.5$*. So the equation of the line goes through points **(0, 13)** and **(-6.5, 0)**.

X	Y
0	13
1	15
2	17
3	19
…	…
7	27



Example 2

Determine the equation of the line through the center (0, 0) and point (3, 5)!

Answer:

y - b = m (x - a)

$$m = \frac{y - b}{x - a} \text{ or } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{5 - 0}{3 - 0} = \frac{5}{3}$$

Point (0,0)

y - b = m (x - a)

y – 0 = 5/3(x - 0)

Y = 5/3x

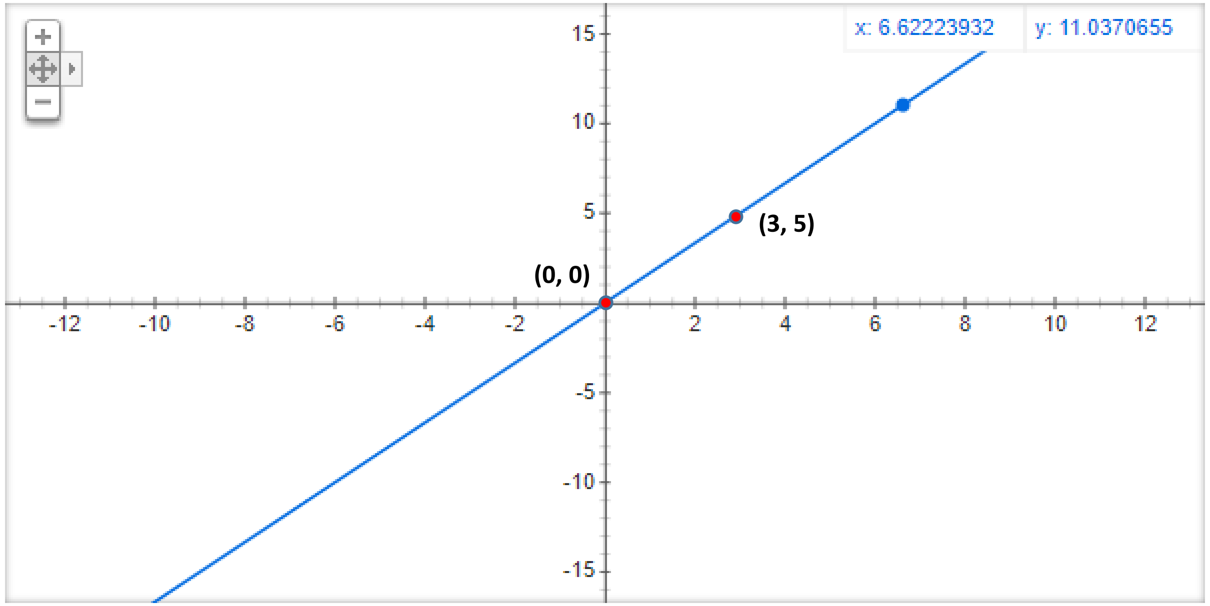
Point (3,5)

y - b = m (x - a)

y – 5 = 5/3(x - 3)

y – 5 = 5/3x – 5

y = 5/3x



Exercise

1. Determine the equation of the line through points (3, 5) and (-2, 4)!
2. Determine the equation of the line through points (-4, 2) and (5, 2)!

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Answer no. 1

Determine the equation of the line through points (3, 5) and (-2, 4)!

Answer:

y - b = m (x - a)

$$m = \frac{y-b}{x-a} \ or \ m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4-5}{-2-3} = \frac{1}{5}$$

Point (3,5)

y - b = m (x - a)

y – 5 = 1/5(x - 3)

Y – 5 = 1/5x – 3/5

Y = 1/5x + 22/5

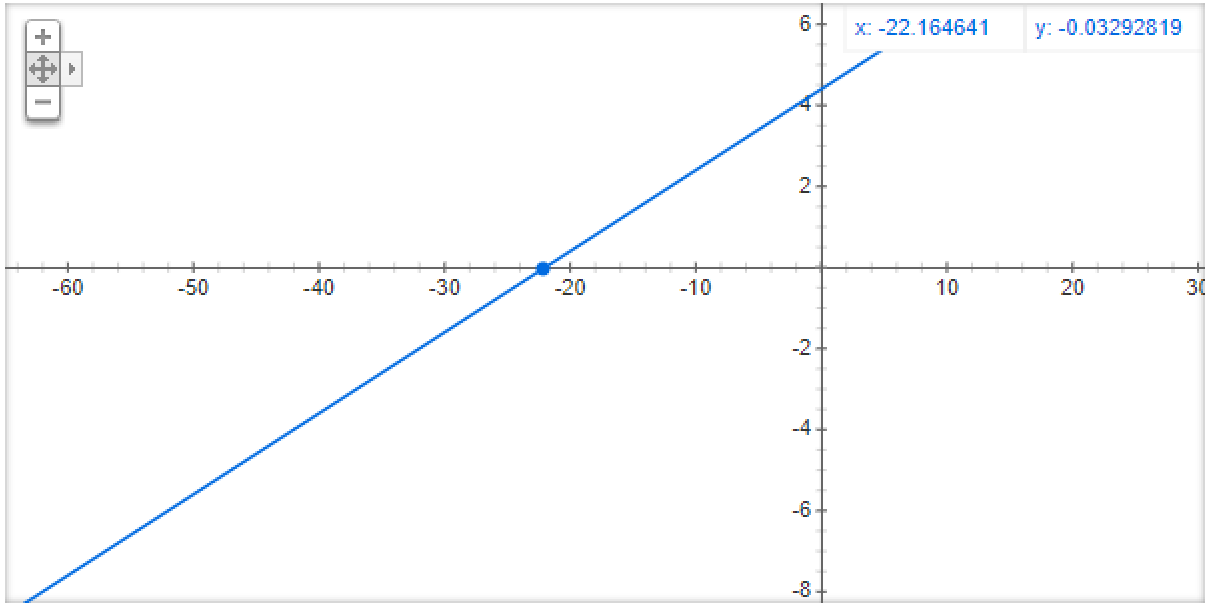
Point (-2,4)

y - b = m (x - a)

y – 4 = 1/5(x + 2)

y – 4 = 1/5x + 2/5

y = 1/5x + 22/5



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Answer no. 2

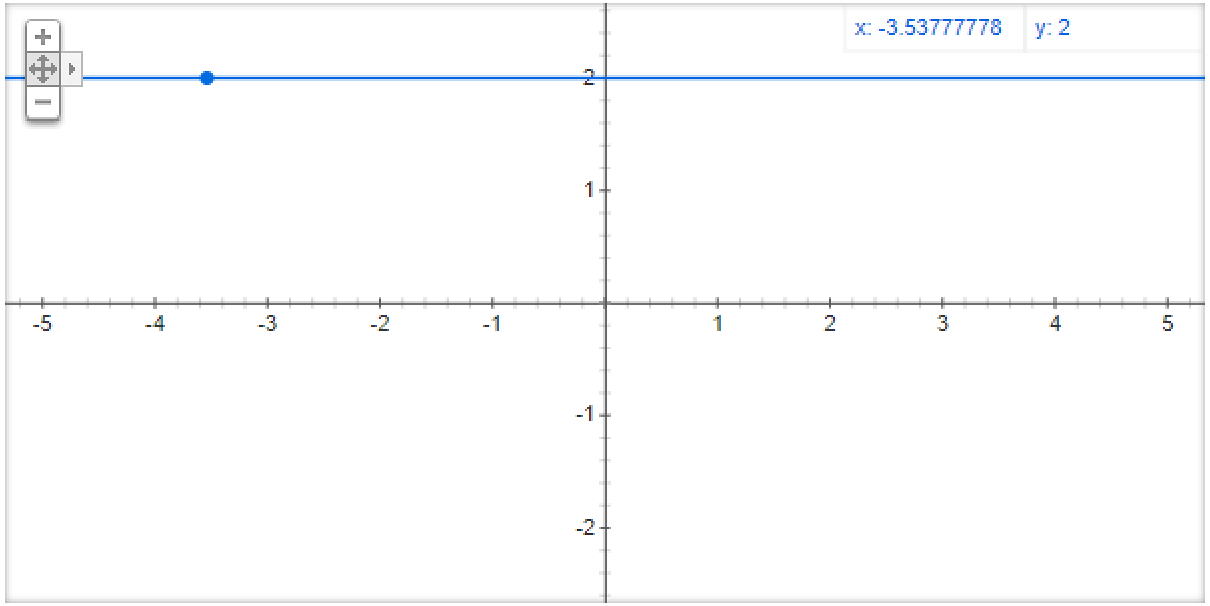
Determine the equation of the line through points (-4, 2) and (5, 2)!

Answer:

y - b = m (x - a)

$$m = \frac{y-b}{x-a} \; or \; m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{2-2}{-5+9} = \frac{0}{9} = 0$$



Point (-4,2)

y - b = m (x - a)

y – 2 = 0(x + 4)

Y = 2

Point (5,2)

y - b = m (x - a)

y – 2 = 0(x - 5)

y = 2

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Perpendicular Line

The gradient of two lines perpendicular to each other if multiplied by the result is equal to -1. So, if line "*l*" is perpendicular to line "*p*" then it applies $\boldsymbol{ml \times mp = -1}$.

Example 1

The line “k” has the equation $y = 2x + 5$. If line “l” is perpendicular to line “k”, determine the gradient of line “l” !

Answer:

$ml \quad = 2$

$mk \times ml \quad = -1$

$\quad = -1/2$

So, the line gradient of “l” is **– 1/2**

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Example 2

Determine the equation of the line through the point (-2, 4) and perpendicular to the line $y = 2x - 4$!

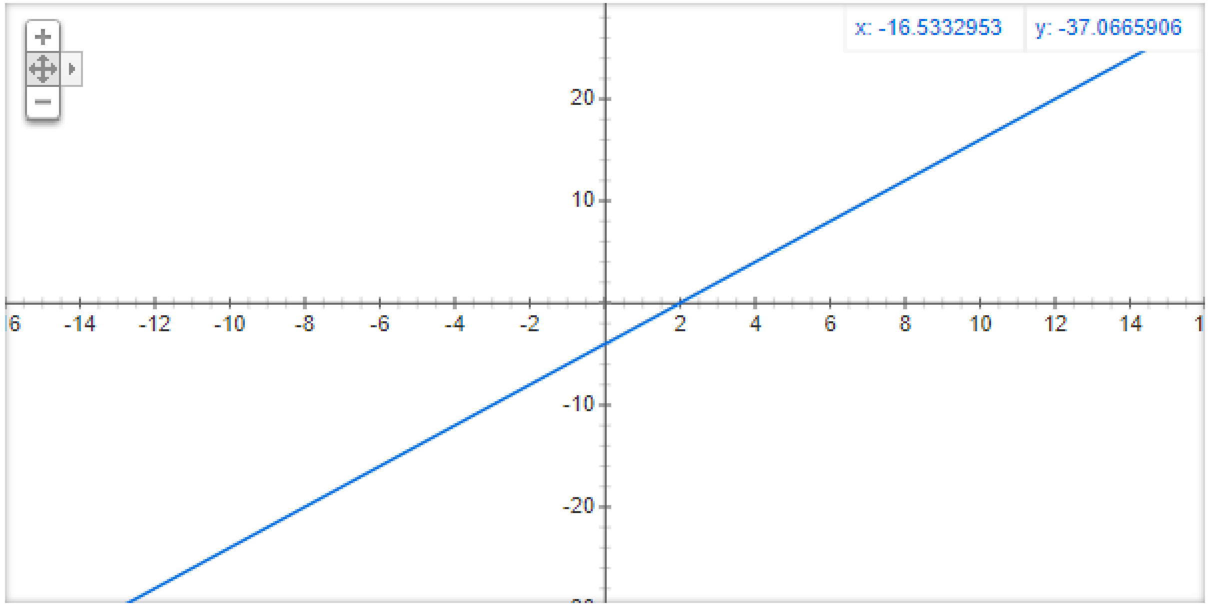
Answer:

The gradient line $y = 2x - 4$ is $m = 2$

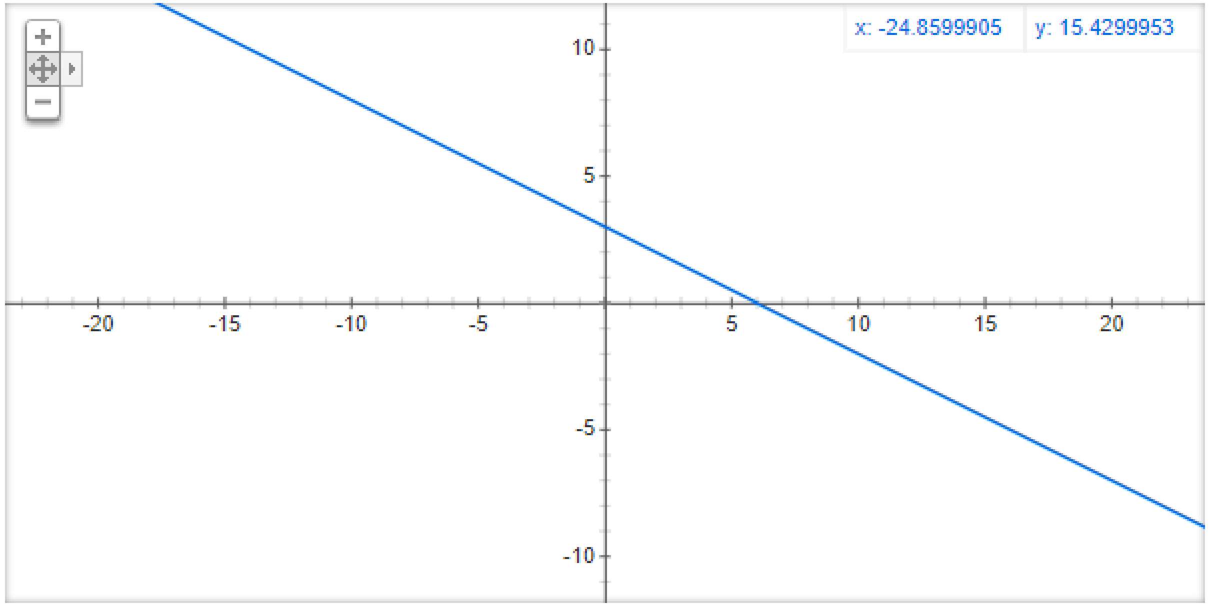
$y - b = -1/m\ (x - a) \Leftrightarrow y - 4 = -1/2\ (x - (-2))$

$\Leftrightarrow y - 4 = -1/2x - 1$

$\Leftrightarrow y = -1/2x + 3$



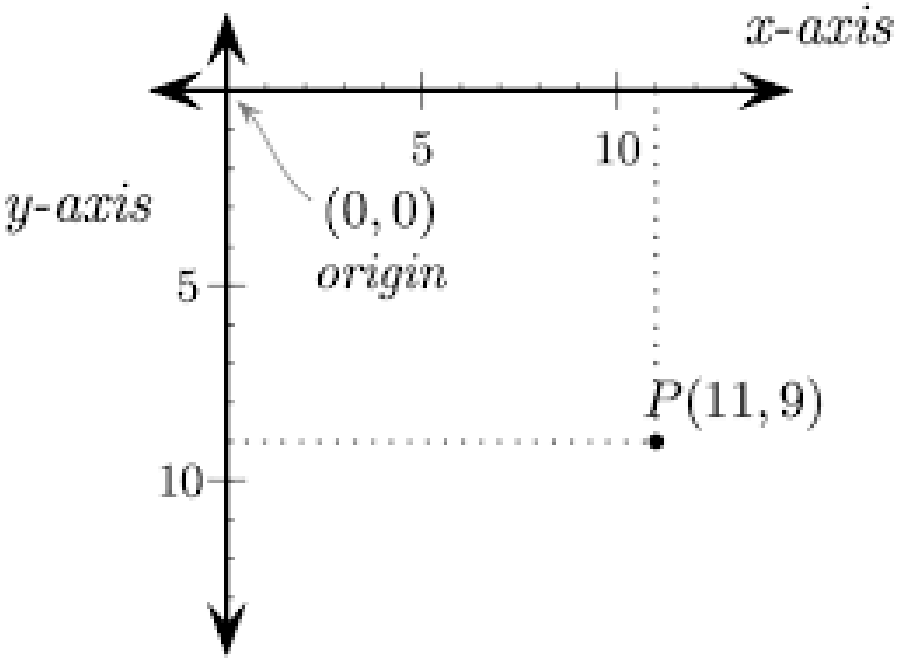
$y = 2x - 4$



$y = -1/2x + 3$

2. Digital Differential Analyzer (DDA)

For using graphics functions, output screen system is treated as a coordinate system where the coordinate of the top-left corner is (0, 0) and as we move down our x-ordinate increases and as we move right our y-ordinate increases for any point (x, y).

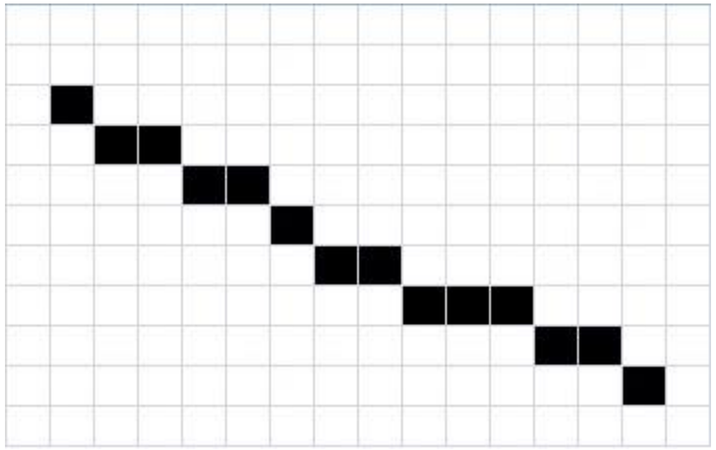


In any 2-Dimensional plane if we connect two points **(x0, y0)** and **(x1, y1)**, we get a line segment.

But in the case of computer graphics (raster) we can not directly join any two coordinate points, for that we should calculate intermediate point's coordinate and put a pixel for each intermediate point, of the desired color with help of functions like put pixel **(x, y, K)** in **C**, where **(x, y)** is our co-ordinate and **K** denotes color and **C** is a canvas.

“Good” Discrete Line

- No gaps in adjacent pixels
- Pixels close to ideal line
- Consistent choices; same pixels in same situations
- Smooth looking
- Lines should have constant density
- Same line for P_0P_1 as for P_1P_0



Input: For line segment between (2, 2) and **(6, 6)** ;

we need (3, 3) (4, 4) and (5, 5) as our intermediate points.

Input: For line segment between (0, 2) and **(0, 6)** ;

we need (0, 3) (0, 4) and (0, 5) as our intermediate points.

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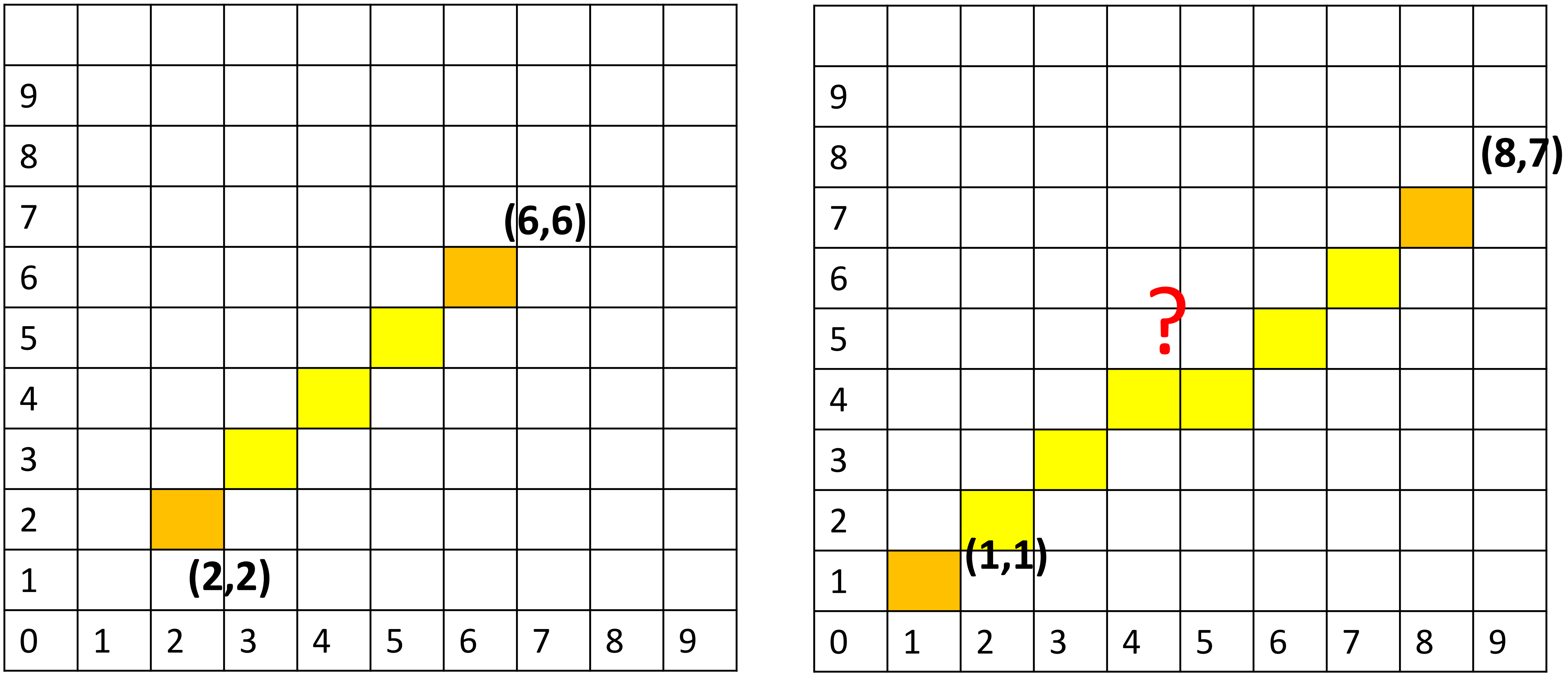


Active Pixel (Intermediate Points)

Digital Differential Analyzer (DDA)

DDA is a line formation algorithm based on the calculation of Δx and Δy, using the formula $\Delta y = m\Delta x$.

The line is created by specifying two endpoints, namely starting point and end point.

Steps for formation according to the DDA algorithm:

1. Determine the two points will be connected.
2. Determine one point as the starting point (x0, y0) and end point (x1, y1).
3. Calculate $\Delta x = x1 - x0$, and $\Delta y = y1 - y0$.
4. Determine the "step", which is the maximum of the $|\Delta x|$ or $|\Delta y|$.
   a) If the value of $|\Delta y| > |\Delta x|$, then "step" = $|\Delta y|$
   b) If not then "step" = $|\Delta x|$
5. Calculate pixel coordinates, x_increment = Δx / step and y_increment = Δy / step.
6. The next coordinate is (x + x_increment, y + y_increment).
7. The pixel position at the layer is determined by rounding the coordination value.
8. Repeat steps 6 and 7 to determine the next pixel position, until x = x1 and y = y1.

Step 1-2

To describe the DDA algorithm in the formation of a line connect the dots **A(10, 10)** and **B(17, 16)**.

Step 3

First determine *dx* and *dy*, then look for steps to get *x_increment* and *y_increment*.

$\Delta x = x1 - x0 = 17 - 10 = 7$ → $|\Delta x| = 7$

$\Delta y = y1 - y0 = 16 - 10 = 6$ → $|\Delta y| = 6$

Step 4

Because $|\Delta x| > |\Delta y|$ is fulfilled, then "step" is $|\Delta x| = 7$

Step 5

x_increment = 7/7 = 1

y_increment = 6/7 = 0,86

Step 6-7

x + x_increment → 10 + 1 = 11

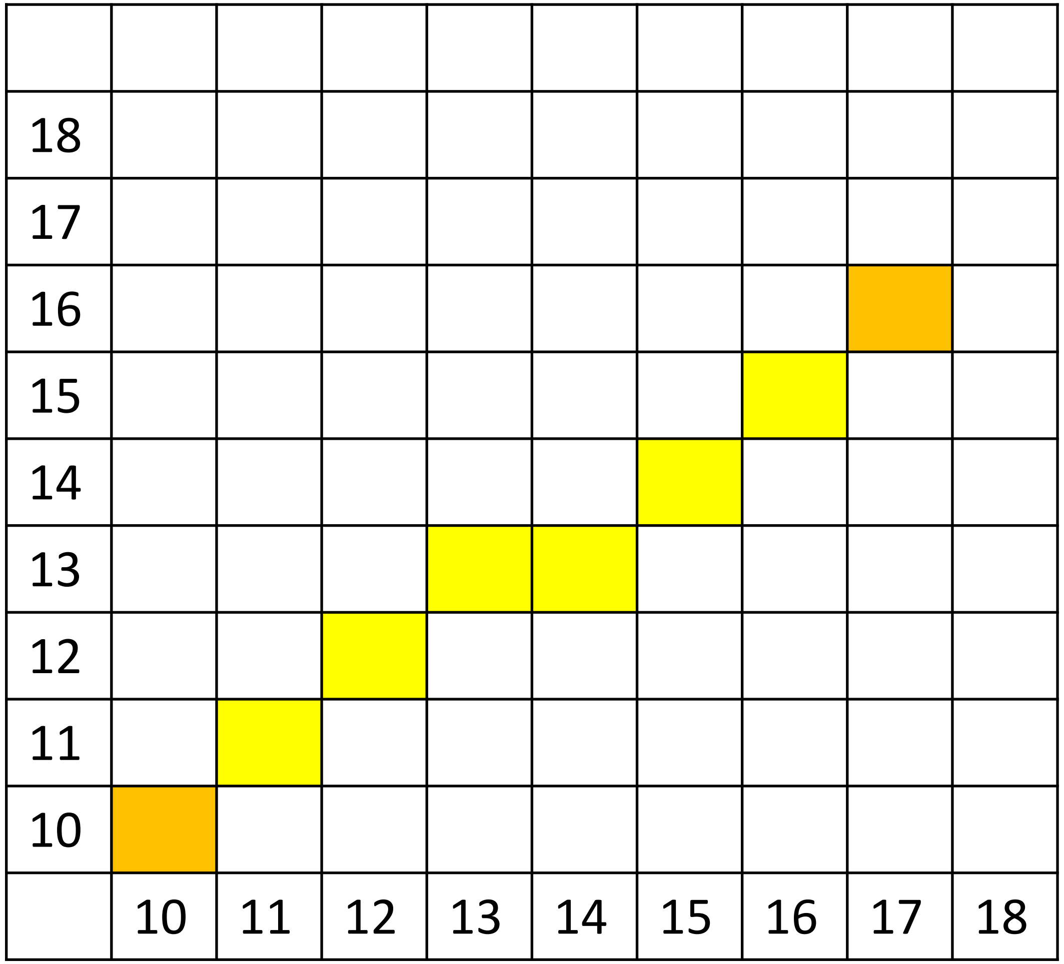
y + y_increment → 10 + 0,86 = 10,86 = 11

Step 8 *(repeat from step 6-7)*

x + x_increment → 11 + 1 = 12

y + y_increment → 10,86 + 0,86 = 11,72 = 12

k	x	y	Round(x), Round(y)
			(10,10)
0	11	10.86	(11,11)
1	12	11.72	(12,12)
2	13	12.58	(13,13)
3	14	13.44	(14,13)
4	15	14.3	(15,14)
5	16	15.16	(16,15)
6	17	16.02	(17,16)



Advantages of DDA algorithm

1. It is a faster method than method of using direct use of line equation.
2. This method does not use multiplication theorem.
3. It allows us to detect the change in the value of x and y ,so plotting of same point twice is not possible.
4. This method gives overflow indication when a point is repositioned.
5. It is an easy method because each step involves just two additions.

Disadvantages of DDA algorithm

1. Floating point arithmetic in DDA algorithm is still time consuming.
2. The algorithm is orientation dependent. Hence end point accuracy is poor.
3. Although DDA is fast, the accumulation of round-off error in successive additions of floating point increment, however can cause the calculation pixel position to drift away from the true line path for long line segment.

THANKYOU